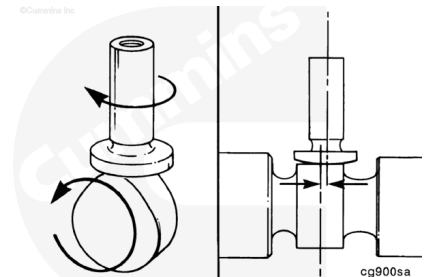


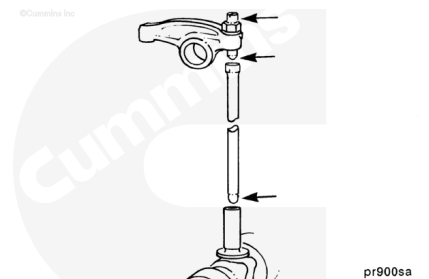
Trainings ▾

General Information

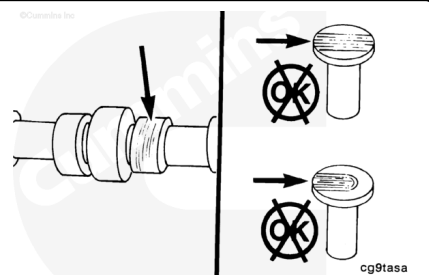
The valve tappets are mushroom shaped. The offset position of the tappet against the camshaft lobe causes the tappet to rotate as it lifts the push rod.



The ball end of the push rod fits into the ball socket in the tappet. The other end of the push rod has a ball socket in which the ball end of the rocker lever adjusting screw operates.

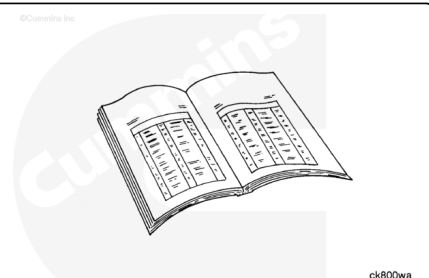


Loose rocker levers and the need to reset the valve clearance frequently can indicate camshaft lobe or tappet wear. If an inspection of the levers, valve stems and push rods does **not** show wear, then tappet or camshaft lobe wear can be suspected.



Preparatory Steps

- Remove the crankcase breather. Refer to Procedure 003-018 (</qs3/pubsys2/xml/en/procedures/100/100-003-018.html>).
- Remove the rocker lever cover. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/100/100-003-011.html>).



- Remove the rocker levers. Refer to Procedure 003-008 (/qs3/pubsys2/xml/en/procedures/100/100-003-008.html).
- Remove the push rods. Refer to Procedure 004-014 (/qs3/pubsys2/xml/en/procedures/100/100-004-014.html).
- Remove the crankshaft pulley/vibration damper. Refer to Procedure 001-052 (/qs3/pubsys2/xml/en/procedures/100/100-001-052.html).
- Remove the front gear cover. Refer to Procedure 001-031 (/qs3/pubsys2/xml/en/procedures/100/100-001-031.html).

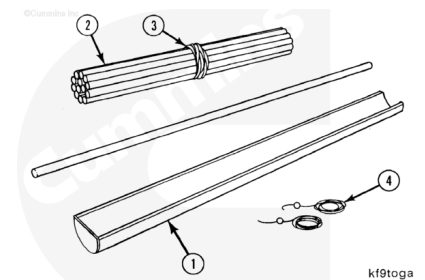
Remove

Oil Pan Installed

The valve tappets can be removed with the oil pan installed by using tappet removal tool kit, Part Number 3822513.

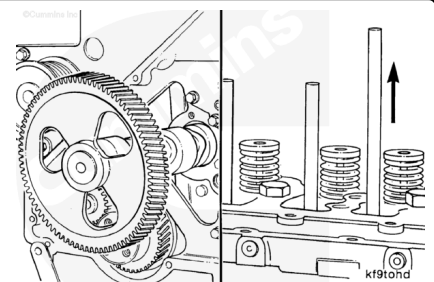
The kit consists of:

1. Valve Tappet Tray - Quantity 1
2. Dowel Rods - Quantity 12
3. Rubber Bands - Quantity 12
4. Nylon String - Quantity 1.

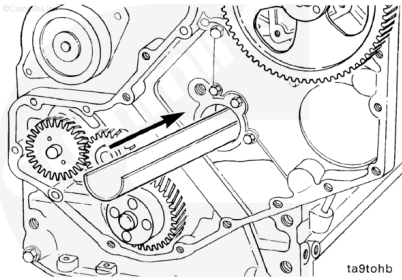


Note : Number each tappet with the cylinder number position as it is removed. The tappets must be installed in the same position as removed.

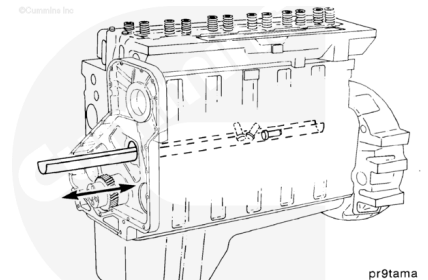
Remove the camshaft. For C Series diesel engines, refer to Procedure 001-008 (/qs3/pubsys2/xml/en/procedures/39/39-001-008.html) in Troubleshooting and Repair Manual, C Series Engines, Bulletin 3666003. For C Series gas engines, refer to Procedure 001-008 (/qs3/pubsys2/xml/en/procedures/41/41-001-008.html) in Troubleshooting and Repair Manual, C8.3G (Natural Gas) Engines, Bulletin 3666206.



Insert the valve tappet tray.

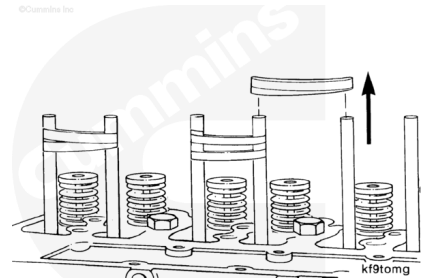


Make sure the valve tappet tray is positioned so it will catch the tappet when the wooden dowels are removed.

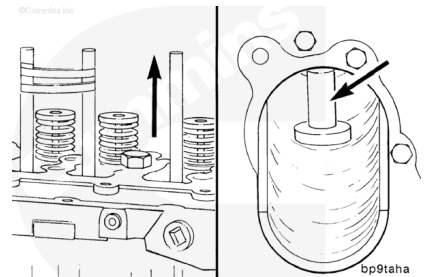


Only remove one tappet at a time.

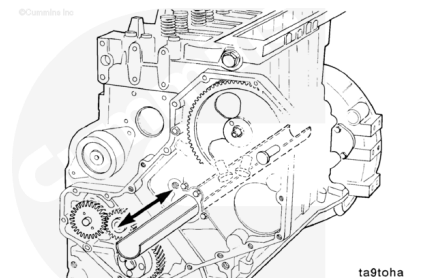
Remove the rubber band from the two companion tappets, securing the tappet **not** to be removed with the rubber band.



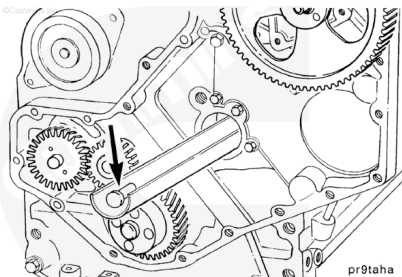
Pull the wooden dowel from the tappet bore, allowing the tappet to fall into the plastic tray.



Note : The tappet must be in the horizontal position to remove it from the camshaft bore. When the tappet is dropped into the tray, it will fall into the horizontal position. If it does not, gently shake the tray to allow the tappet to fall over.



Carefully pull the tray and tappet from the cam bore and remove the tappet. Repeat the process until all tappets are removed.



Oil Pan Removed

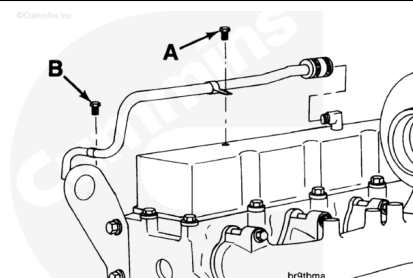
⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

Remove the oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/100/100-007-025.html>).

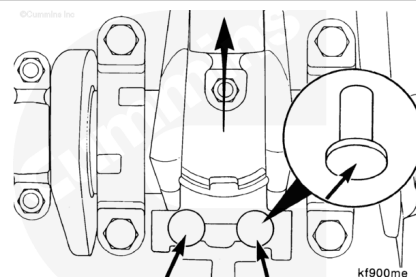


Rotate the crankshaft with engine barring gear, Part Number 3377371, to access the tappets.

Note : Only four tappets are accessible at one time.

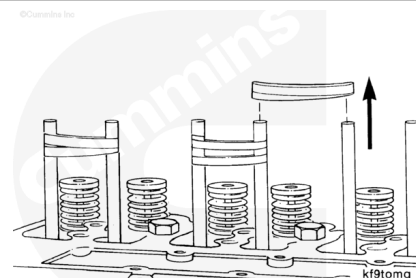
The crankshaft **must** be rotated three times to access all of the tappets.

Note : Number each tappet with the cylinder number position as it is removed. The tappets must be installed in the same position as removed.

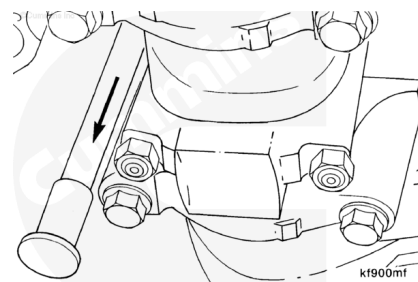


Remove the rubber bands from the wooden dowels that are used to hold the tappets up during camshaft removal.

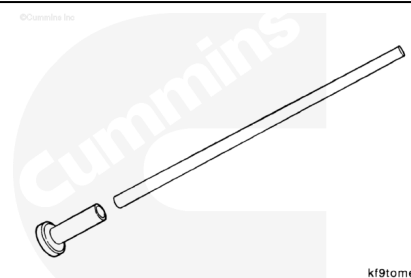
Note : Only remove the rubber bands from the accessible tappets.



Remove the accessible tappets and dowels.



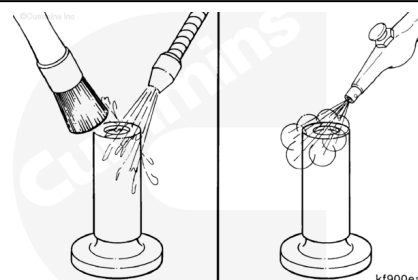
Remove the tappets from the wooden dowels.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the tappets with solvent.

Dry the tappets with compressed air.

⚠ CAUTION ⚠

Anytime a new camshaft is installed, new tappets and push rods must also be installed.

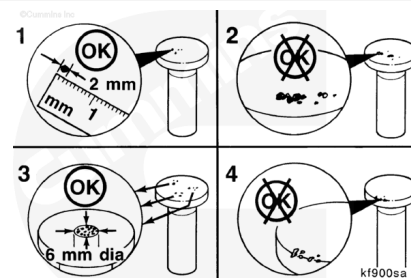
⚠ CAUTION ⚠

If a tappet is replaced, the corresponding push rods must also be replaced. Failure to do this can cause major engine damage.

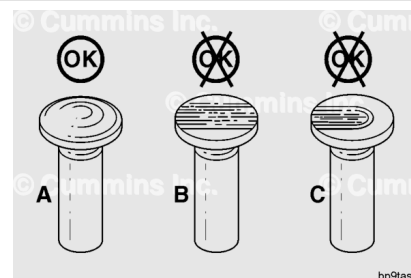
Pit marks on the tappet face are acceptable.

The following criteria define the size of the pits allowed:

1. A single pit can **not** be greater than 2 mm [0.078 in].
2. Interconnection of pits is **not** allowed.
3. Total pits added together **must not** exceed 6-mm [0.236-in] diameter or a total of 4 percent of the tappet face.
4. No pitting is allowed on the edges of the wear face of the tappet.



Inspect the socket, stem, and face for excessive wear, cracks, and other damage.



Visual Limits

(A) - Normal Contact (exaggerated)

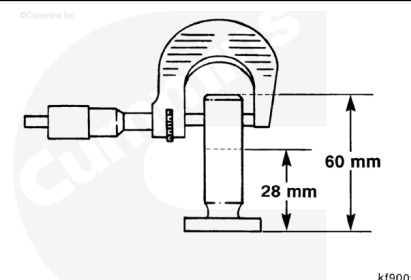
(B) and (C) - Irregular Contact: Do **not** reuse.

Measure the valve tappet stem diameter.

mm		in
15.93	MIN	0.627
15.98	MAX	0.629

Measure the valve tappet height.

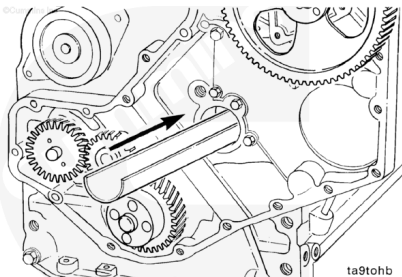
mm		in
60	MAX	2.36



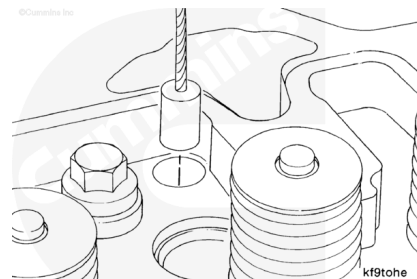
Install

Oil Pan Installed

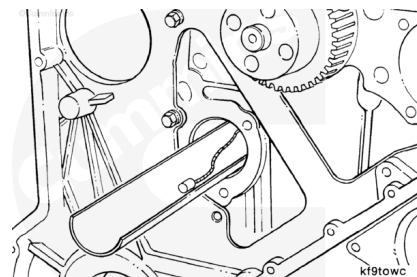
Insert the valve tappet tray the full length of the cam bore.



Lower the tappet installation tool down the push tube hole, through the tappet bore, and into the tray.



Feed the installation tool through the cam bores by carefully pulling the plastic tray/installation tool out the front of the cylinder block.



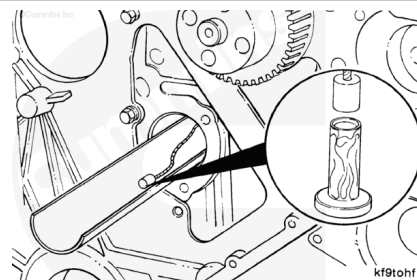
Note : If the tappets to be installed have been previously used, each tappet must be installed in the same position as it was removed from the engine.

Lubricate the tappets with clean engine oil.

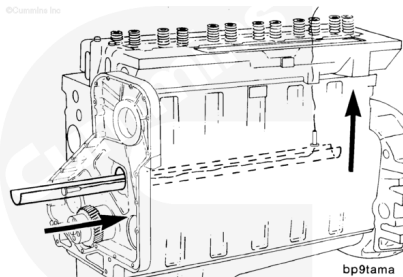


Note : To aid in removing the installation tool after the tappet is installed, work the tool in and out of the tappet several times before installing the tappets.

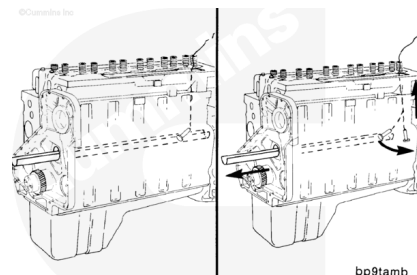
Insert the installation tool into the tappet.



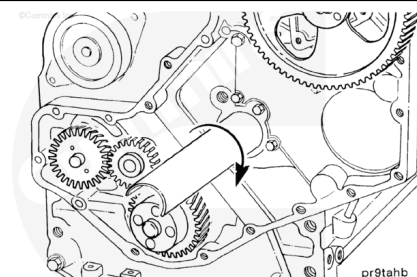
Pull the installation tool and tappet through the cam bore and up into the tappet bore.



Note : If difficulty is experienced in getting the tappet to make the bend from the tray up to the tappet bore, pull the tray out enough to allow the tappet to drop down and align itself; then pull the tappet up into the bore.



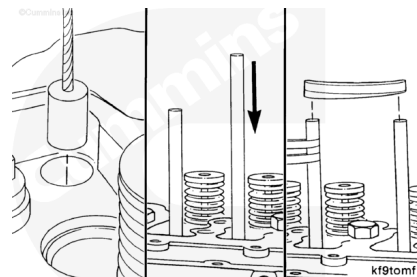
After the tappet has been pulled up into position, slide the tray back into the cam bore and rotate it a $\frac{1}{2}$ turn. This will position the round side of the tray up and hold the tappet in place.



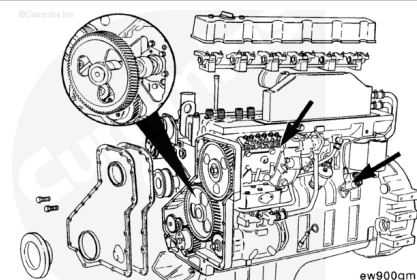
Remove the installation tool from the tappet.

Install a wooden dowel into the top of the tappet. Wrap rubber bands around the wooden dowels to secure the tappets.

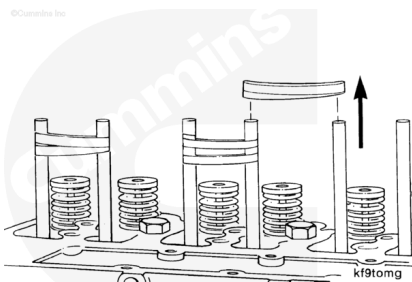
Repeat this process until all tappets have been installed.



Install the camshaft. For C Series diesel engines, refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/39/39-001-008.html>) in Troubleshooting and Repair Manual, C Series Engines, Bulletin 3666003. For C Series gas engines, refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/41/41-001-008.html>) in Troubleshooting and Repair Manual, C8.3G (Natural Gas) Engines, Bulletin 3666206.

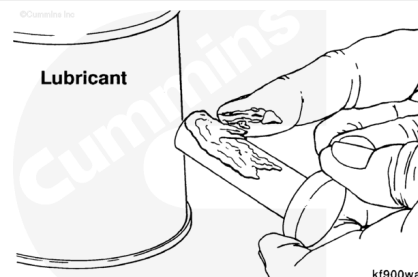


Remove the dowel rods and release the tappets in the tappet bores.

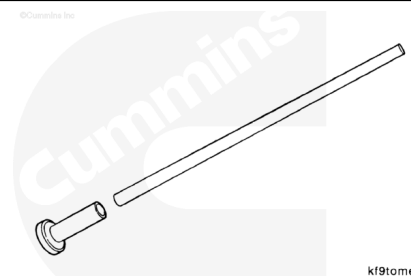


Oil Pan Removed

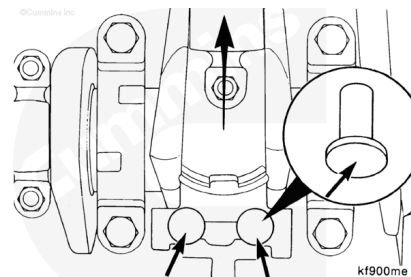
Lubricate the tappets with clean engine oil.



Insert the wooden dowel rods used for holding the tappets up during camshaft installation into the tappets.

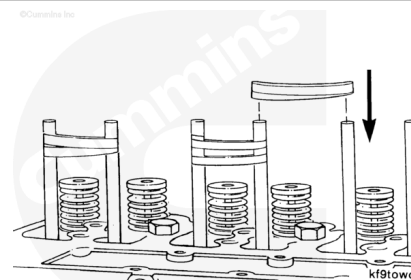


Note : The crankshaft must be rotated during the installation process. Only four tappets can be installed at each crankshaft position.



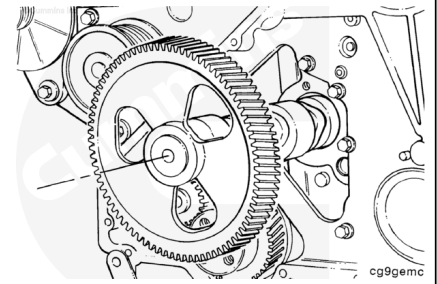
Wrap rubber bands around the wooden dowels to secure the tappets.

Repeat this process until all tappets have been installed.

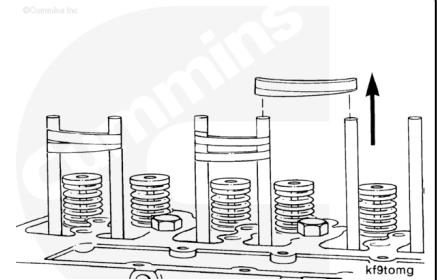


Install the camshaft. For C Series diesel engines, refer to Procedure 001-008 (/qs3/pubsys2/xml/en/procedures/39/39-001-008.html) in Troubleshooting and Repair Manual, C Series

Engines, Bulletin 3666003. For C Series gas engines, refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/41/41-001-008.html>) in Troubleshooting and Repair Manual, C8.3G (Natural Gas) Engines, Bulletin 3666206.



Remove the dowel rods and release the tappets in the tappet bores.



Finishing Steps

- Install the oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/100/100-007-025.html>).
- Fill the engine with clean engine oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/100/100-007-037.html>).
- Install the front gear cover. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/100/100-001-031.html>).
- Install the push rods. Refer to Procedure 004-014 (</qs3/pubsys2/xml/en/procedures/100/100-004-014.html>).
- Install the rocker levers. Refer to Procedure 003-008 (</qs3/pubsys2/xml/en/procedures/100/100-003-008.html>).
- Adjust the overhead set. For C Series diesel engines, refer to Procedure 003-004 (</qs3/pubsys2/xml/en/procedures/39/39-003-004.html>) in Troubleshooting and Repair Manual, C Series Engines, Bulletin 3666003. For C Series gas engines, refer to Procedure 003-004 (</qs3/pubsys2/xml/en/procedures/41/41-003-004.html>) in Troubleshooting and Repair Manual, C8.3G (Natural Gas) Engines, Bulletin 3666206.



- Install the rocker lever cover. Refer to Procedure 003-011 (/qs3/pubsys2/xml/en/procedures/100/100-003-011.html).

Operate the engine until the coolant temperature reaches 82°C [180°F], and check for leaks and proper operation.

Last Modified: 22-Sep-2003
